Experimental Procedure

Prototype testing and data collection plan

The testing plan addresses all or nearly all of the high priority design requirements by effectively describing the conduct (through physical and/or mathematical modeling) of those tests that are feasible based on the instructional context and providing for others a logical and well-developed explanation confirmed by one or more field experts of how testing would yield objective data regarding the effectiveness of the design.

Brainstorming Phase

When we first sat down to brainstorm ideas for the project, we had no shortage of thoughts on how to fix the problems that had come up in the earlier prototypes. Everyone brought their own perspective to the table, and we spent a lot of time tossing ideas back and forth, trying to figure out what went wrong before and how we could make things better this time around.

Focusing on Strength

One of the biggest issues we noticed in the older designs was that they just weren't strong enough. The frame would start to bend or warp if anything over 150 pounds was placed in the sling. That obviously wasn't ideal—especially for the vet techs, who had to work with it directly and felt the consequences of that instability. We knew right away that structural strength had to be a top priority.

Our first thought was pretty straightforward: add more support beams and improve how the parts were joined together. It wasn't just about adding more material—it was about placing it where it would actually help carry the load better. We started sketching out ideas and comparing what might work best, always keeping in mind that it still had to be practical and safe to use.

Testing with Popsicle Sticks

At one point, Corbin—who was helping us with the building side of things—mentioned that welding was on the table. That opened up a lot of new possibilities, but it also meant we had to give him a clear design to work with. That's what led us to our popsicle stick phase.

We built small mock-ups using popsicle sticks and glue to get a feel for different frame layouts. It might sound a little elementary, but it was actually a great way to quickly see what held up and what didn't. We tested different shapes and support placements, paying attention to where the models flexed or stayed solid. These simple tests gave us a much better understanding of how force would move through the frame—and helped us decide on a final design that we felt confident could handle real weight without buckling.

Each of the frames were made using nothing but popsicle sticks and hot glue as our "welds". The tests we performed were nothing impressive, but gave a good look into strengths and weaknesses of each format.

The frames had strings tied to each corner and in the middle on each side, where at the bottom was a duct tape tarp which was used to hold marbles, as weight. They were attached to a pole above using the same type of string, faceted at each corner.

(A brief description of each frame in order of the picture)

- The “Ribcage”
    - Used moderate amount of resources while being fortified enough to handle 36 marbles
- The “ZigZag”
    - Used a little over the required resources, yet stretched out weekpoints and had even strength around the perimeter of the frame
    - 27 marbles
- The X
    - Used the least amount of resources, which was great for budget, but ws by far the weakest
    - 18 marbles
- The “Coffin”

- Used the most of resources, but made up for it in strength and fortification
- 55 Marbles

Stability

Using the popsicle stick frames to determine which fortification format would work best, we constructed a table-top prototype out of cardboard, popsicle sticks, and string. Nic originally built it as a way to test weakpoint, but then made the stick frames to test strength, so he used the larger frame to test different ways we could orient the frame to allow for maximum adaptation.

In his test that he conducted, he tied string to the frame in the same orientation as the smaller frames, and held the strings that we pointed up in order to act as an adaptable factor in teh frames stability.

The strings fed down to a piece of cardboard that was used to carry a hammer, which we used to imitate an animal with uneven weight distribution throughout the body. When we placed the hammer in the middle of the cardboard plate, the frame didn't tip at all. It was when we placed the hammer on the edge when we realized why stability was such an issue. The frame tilted about 25° from the horizontal, causing the hammer to roll off the plate. Our goal was to be able to keep the hammer on the plate wherever it was placed.

We ended up creating a system that shifted the animal or hammer, wherever it was on the plate, to right underneath that points of contact with the “rail”. This was tested by, with each string in hand, switching from hanging it on Nic’s index finger all the way to his pinky finger, shifting the actual position the frame was below his arm, instead of just tilting the frame.